

A Repaired Split-Forest Automata Proof of Decidability for

$\mathcal{ALCI}_{\text{RCC5}}$

Patchwork Bags, Exact Occurrence Identity, and Two-Way Parity Checking

Prepared as a consolidated proof manuscript

May 31, 2026

Abstract

This note gives a repaired proof scheme for decidability of concept satisfiability for $\mathcal{ALCI}$ whose roles are the five atomic RCC5 relations EQ, PP, PPI, PO, DR, interpreted over abstract complete atomic RCC5 frames with strong equality. The proof keeps the split-forest idea as the semantic normal form, but it replaces the earlier transformer-based finite abstraction by a patchwork-bag abstraction. The key change is that global RCC5 coherence is no longer derived from route-dependent pair and triple transformers. Instead, each bag carries a finite complete RCC5 network, adjacent bags agree on overlaps, and global consistency follows from the patchwork property of RCC5. Equality is treated as exact occurrence identity in the tree decomposition, not as a quotient of repeated blocking profiles. Universal restrictions are checked by relation-set shadows, and infinite vertical eventualities are handled by a two-way alternating parity tree automaton. This is a proof for abstract composition-table RCC5 frames, not for topological RCC5 semantics.

Contents

1	Motivation and relation to the original split-forest idea	2
2	RCC5 frames and the two external finite-model facts	2
2.1	The abstract frame semantics	2
2.2	Patchwork for complete RCC5 networks	3
2.3	Two-way alternating parity tree automata	4
3	Syntax, closure, types, and witness thinning	4
4	Saturated split-forest presentations	4
5	Finite patchwork abstraction	6
5.1	Shadows and universal obligations	6
5.2	Patchwork abstraction	7
6	The automaton	9
7	Decidability	10

8	Expanded construction details	10
8.1	From a generated model to bags	10
8.2	The finite alphabet	11
8.3	Overlap and running-intersection checks	11
8.4	The patchwork construction in detail	12
8.5	Universal propagation by exact relation vectors	12
8.6	Vertical eventualities and equality transfer	12
9	Audit against the referee concerns	13
10	Limitations and scope	13

1 Motivation and relation to the original split-forest idea

The original split-forest idea remains the essential model-theoretic normal form. Its purpose is not to compute the Hasse diagram of an arbitrary semantic RCC5 model. Rather, it says that after witness thinning one may work with a generated containment presentation: selected PP/PPI witnesses are represented vertically, selected DR/PO witnesses are represented in finite side contexts, and multiple incomparable proper-superpart witnesses are handled by split occurrences connected through explicit identity ports.

The part that has changed is the finite abstraction layer. Earlier drafts attempted to recover the global relation $\rho(x, y)$ by pushing pair and triple states through interface transformers. That creates a coherence problem: computing a pair from its least common interface and computing the same pair as a projection of a higher triple need not agree unless a nontrivial confluence theorem is proved. The present proof avoids that issue. The finite object is a tree decomposition by *patchwork bags*. Each bag is a finite complete RCC5 network. The only required compatibility between bags is agreement on shared occurrences. Global RCC5 consistency then follows from the RCC5 patchwork theorem. Thus the split forest supplies the normal form, while the automaton checks existence of a regular labelled tree of finite patchwork bags.

This gives the following theorem spine.

Theorem A. Every satisfiable concept has a witness-generated saturated split-forest presentation.

Theorem B. Saturated split forests are exactly represented, up to the closure of the input concept, by finite patchwork-bag abstractions.

Theorem C. A finite two-way alternating parity tree automaton accepts exactly the patchwork-bag abstractions satisfying the input concept.

The decidability theorem follows from the decidability of emptiness for finite two-way alternating parity tree automata.

2 RCC5 frames and the two external finite-model facts

2.1 The abstract frame semantics

Let

$$\text{Rel} = \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}, \text{DR}\}.$$

An *abstract complete atomic RCC5 frame* is a nonempty set Δ together with a map

$$\rho : \Delta^2 \rightarrow \text{Rel}$$

such that for all $x, y, z \in \Delta$:

- (R1) $\rho(x, y) = \text{EQ}$ iff $x = y$;
- (R2) $\rho(y, x) = \rho(x, y)^{-1}$, where $\text{PP}^{-1} = \text{PPI}$, $\text{PPI}^{-1} = \text{PP}$, and $\text{EQ}, \text{PO}, \text{DR}$ are self-converse;
- (R3) $\rho(x, z) \in \text{Comp}(\rho(x, y), \rho(y, z))$, where Comp is the RCC5 composition operation.

We use the standard RCC5 composition table:

Comp	EQ	PP	PPI	PO	DR
EQ	{EQ}	{PP}	{PPI}	{PO}	{DR}
PP	{PP}	{PP}	Rel	{PP, PO, DR}	{DR}
PPI	{PPI}	{EQ, PP, PPI, PO}	{PPI}	{PPI, PO}	{PPI, PO, DR}
PO	{PO}	{PP, PO}	{PPI, PO, DR}	Rel	{PPI, PO, DR}
DR	{DR}	{PP, PO, DR}	{DR}	{PP, PO, DR}	Rel

An interpretation of $\mathcal{ALCI}_{\text{RCC5}}$ consists of such a frame and an assignment of concept names to subsets of Δ . The five RCC5 relations are ordinary roles:

$$\mathcal{I}, x \models \exists R.C \quad \text{iff} \quad \text{there is } y \in \Delta \text{ with } \rho(x, y) = R \text{ and } \mathcal{I}, y \models C,$$

and similarly for universal restrictions. Since EQ is identity, $\exists \text{EQ}.C$ and $\forall \text{EQ}.C$ are equivalent to C ; below we normalize them away.

2.2 Patchwork for complete RCC5 networks

The proof uses one algebraic fact about RCC5.

Theorem 2.1 (RCC5 patchwork property). *Let N_1, N_2 be finite complete atomic RCC5 networks over variable sets V_1, V_2 . Assume that each N_i is satisfiable as an abstract RCC5 network and that N_1 and N_2 agree on the common variables $V_1 \cap V_2$. Then the union network on $V_1 \cup V_2$ is satisfiable. The same holds for any finite tree-shaped family of finite complete RCC5 networks agreeing on pairwise overlaps. Moreover, any countable tree-shaped family has a countable abstract completion all of whose finite restrictions are patchwork unions of finitely many bags.*

Proof indication. The two-network statement is the usual patchwork property of atomic RCC5. The finite family statement follows by induction over the tree of bags, repeatedly amalgamating one leaf bag with the already amalgamated remainder. For a countable tree of bags, every finite set of variables is contained in the union of finitely many bags after adding the finite connecting subtree. That finite restriction is satisfiable by the finite patchwork property. Since the relation alphabet is finite, a standard compactness/Konig argument yields a labelling of all pairs in the countable union whose every finite restriction is one of these satisfiable finite patchworks. This is an abstract composition-table argument; it does not claim topological representability. $\square$

Theorem 2.1 is precisely what replaces the earlier transformer-coherence lemma. No relation in the global model is computed by following an interface route. It is supplied by a patchwork completion of locally consistent complete bag networks.

2.3 Two-way alternating parity tree automata

We also use the standard fact that emptiness for finite two-way alternating parity tree automata over finite ranked alphabets is decidable. Two-way motion is convenient because it allows the automaton to revisit ancestor and sibling interfaces instead of relying on lossless one-way shadows. The use of two-way motion is only in the decision layer; the semantic normal form remains a split-forest presentation.

3 Syntax, closure, types, and witness thinning

Put all concepts in negation normal form. For each NNF concept D , write $\overline{D}$ for its NNF complement. Let $\text{Cl}(C_0)$ be the least finite set containing C_0 , closed under subformulas, under $D \mapsto \overline{D}$, and with $\exists \text{EQ}.D, \forall \text{EQ}.D$ replaced by D .

Definition 3.1 (Hintikka type). *A type over $\text{Cl}(C_0)$ is a set $t \subseteq \text{Cl}(C_0)$ such that:*

- (T1) *exactly one of $D, \overline{D}$ is in t for each $D \in \text{Cl}(C_0)$;*
- (T2) *$D \sqcap E \in t$ iff $D, E \in t$, and $D \sqcup E \in t$ iff $D \in t$ or $E \in t$;*
- (T3) *the EQ-normalized clauses $\exists \text{EQ}.D \leftrightarrow D$ and $\forall \text{EQ}.D \leftrightarrow D$ are respected when such formulas occur before normalization.*

Let Typ be the finite set of all such types.

Lemma 3.2 (Witness-generated submodel). *If $\mathcal{I}, a_0 \models C_0$, then there is a subinterpretation $\mathcal{I} \upharpoonright W$ with $a_0 \in W$ such that:*

- (i) *W is generated from a_0 by choosing, for every $a \in W$ and every $\exists R.D \in \text{Cl}(C_0)$ true at a , one witness b with $\rho(a, b) = R$ and $\mathcal{I}, b \models D$;*
- (ii) *for every $a \in W$ and every $D \in \text{Cl}(C_0)$,*

$$\mathcal{I}, a \models D \iff \mathcal{I} \upharpoonright W, a \models D.$$

Proof. Let W be the least subset containing a_0 and all selected witnesses for existential closure formulas at already generated elements. The proof is by induction on $D \in \text{Cl}(C_0)$. Atomic and Boolean cases are immediate. For $D = \exists R.E$, a witness in $\mathcal{I}$ is retained by construction. Conversely, any witness in the restricted model is also a witness in $\mathcal{I}$. For $D = \forall R.E$, the forward direction is inherited by restriction. For the reverse direction, suppose $a \not\models_{\mathcal{I}} \forall R.E$. Then $a \models_{\mathcal{I}} \exists R.\overline{E}$. Since $\text{Cl}(C_0)$ is closed under NNF complement, the selected witness for $\exists R.\overline{E}$ lies in W , and by the induction hypothesis it still satisfies $\overline{E}$ in the restricted model. Hence the universal fails there as well. $\square$

4 Saturated split-forest presentations

The split-forest presentation is a normal form for witness-generated models. It is not a Hasse diagram of an arbitrary original model. It is a generated containment presentation.

Definition 4.1 (Generated containment skeleton). *A generated containment skeleton is a forest-like directed structure whose vertical edges are selected PPI-cover edges downward, equivalently selected PP-cover edges upward. The semantic PP and PPI relations along a branch are the strict ancestor and strict descendant closures of these selected cover edges. The skeleton may have infinite rays. Blocking repetitions are presentation repetitions, not semantic equality.*

A node may need several incomparable selected PP-superpart witnesses. Rather than assigning several parents to one tree node, the presentation uses a tree decomposition of finite bags. The same semantic occurrence may be present in several bags through shared ports; the occurrence is one vertex of the decomposition, not two vertices related by a nontrivial EQ edge.

Definition 4.2 (Patchwork bag). *A patchwork bag is a finite object*

$$B = (P_B, N_B, \tau_B, \iota_B, W_B, U_B, E_B)$$

where:

- (B1) P_B is a finite set of local port positions, drawn from a fixed finite set $\{1, \dots, K\}$;
- (B2) $N_B : P_B^2 \rightarrow \text{Rel}$ is a complete atomic RCC5 network satisfying the composition table on all triples in P_B and with $N_B(p, q) = \text{EQ}$ iff $p = q$;
- (B3) $\tau_B(p) \in \text{Typ}$ is the concept type of p ;
- (B4) ι_B records overlap maps to the parent and children of the bag; overlap maps are partial injections preserving N_B and τ_B ;
- (B5) W_B records local witnesses for existential restrictions, either as a port in the same bag or as a declared vertical request to be fulfilled by parity;
- (B6) U_B records relation-set universal obligations carried by live occurrences and shadows;
- (B7) E_B records exact occurrence identity through overlaps. There is no nontrivial semantic EQ between distinct ports; identity means that two port occurrences are the same vertex of the tree decomposition by the equivalence relation generated by overlap maps.

Definition 4.3 (Saturated side context). *A side context for a source occurrence u is saturated if it is the least finite context closed under the following operations, up to the rank bound described below.*

- (S1) Insert every selected DR or PO witness of u .
- (S2) Insert the vertical boundary of u needed to evaluate relations to its parent, children, selected superparts, selected subparts, and shared identity ports.
- (S3) For every side/tree or side/side pair whose inherited relation is PP, PPI, or EQ, replace the pair by an explicit local vertical stratum or shared identity port. Thus such a pair is never left as an unstructured residual frontier pair.
- (S4) Apply the same operation recursively to side witnesses, with strictly smaller rank measured by the remaining modal depth below the triggering existential formula.
- (S5) Only after this saturation may a residual frontier pair be classified as DR or PO.

Lemma 4.4 (Finite saturation bound). *For fixed C_0 there is a computable number $K(C_0)$ such that every rank- r saturated side context needed for C_0 has a representative using at most $K(C_0)$ local ports, up to equality of types, overlap pattern, finite RCC5 bag network, witness declarations, and universal-obligation summaries.*

Proof. Let $n = |\text{Cl}(C_0)|$ and let $T = |\text{Typ}| \leq 2^n$. For a local context of width at most k , the number of complete atomic RCC5 networks on its ports is at most 5^{k^2} , the number of type assignments is at most T^k , and the number of witness and obligation summaries is bounded by a power set of a set of size polynomial in $k, n, 5$. Hence, for every fixed k , there are finitely many context shapes.

Define K_r inductively as the number of equivalence classes of rank- r context shapes, where rank 0 contains only the source, its finite vertical boundary, and direct witnesses whose targets have no further modal obligations. For rank $r + 1$, a context shape is determined by: the source type; a finite bag network on a live frontier; one direct witness declaration for each existential formula in $\text{Cl}(C_0)$; one threshold verticalization position for each pair of live or side ports whose composition forces PP, PPI, or EQ; and one recursive rank- r subcontext class for each recursive trigger. Since each of these choices ranges over a finite set determined by K_r and n , there are finitely many classes. Let $K(C_0)$ be the maximum size of a chosen representative for all ranks $r \leq n$. This is an effective bound because all classes are enumerable: enumerate all finite labelled bags up to size m , check local RCC5 consistency, saturation closure, overlap compatibility, and increase m until every newly generated rank- r class has a representative. The induction guarantees termination at each rank. No numerical recurrence is used; only effective finiteness of the quotient is needed. $\square$

Theorem 4.5 (A: split-forest normal form). *If C_0 is satisfiable over abstract complete atomic RCC5 frames, then C_0 is satisfiable in a witness-generated saturated split-forest patchwork presentation of width at most $K(C_0)$.*

Proof. Start with a satisfying model and apply Lemma 3.2. Use the selected PP/PPI witnesses to form generated vertical cover links. When one semantic element requires several incomparable selected superparts, introduce several bags sharing the same occurrence through overlap ports rather than identifying distinct copies by a semantic EQ edge. Insert all selected DR/PO witnesses into side contexts and saturate them by (S1)–(S5). The construction is rank-decreasing on recursive side contexts, hence terminates in one of the finitely many context shapes of Lemma 4.4. Each finite bag is labelled by the restriction of the original complete RCC5 network, so every bag is locally satisfiable and adjacent bags agree on their shared occurrences. The resulting presentation is a tree decomposition by finite complete networks, with exact occurrence identity represented by connected overlap classes. $\square$

5 Finite patchwork abstraction

The finite abstraction deliberately avoids pair and triple transformers. Its alphabet contains finite complete bag networks and overlap maps. Global consistency is a patchwork consequence.

5.1 Shadows and universal obligations

A universal restriction at a source occurrence must apply to every target occurrence in the final global network. The automaton can check this with two-way motion, but the bag labels still record finite summaries called shadows.

Definition 5.1 (Shadow profile). *For a bag B , a shadow profile is a tuple*

$$\sigma = (t, \vec{r}, \Omega)$$

where $t \in \text{Typ}$ is the type of the source represented by the shadow, $\vec{r}$ is the vector of RCC5 relations from that source to the live ports of B , and

$$\Omega \subseteq \text{Rel} \times \text{Cl}(C_0)$$

is the set of universal obligations carried by the source: $(R, D) \in \Omega$ means that any target currently known to be in relation R to the source must have D in its type. Two shadows with the same triple impose exactly the same constraints on every extension of the current bag. The finite set of shadow profiles over width K has cardinality at most

$$|\text{Typ}| \cdot 5^K \cdot 2^{5|\text{Cl}(C_0)|}.$$

Definition 5.2 (Relation-set update). *When moving from a bag B to an adjacent bag B' across overlap O , a shadow $(t, \vec{r}, \Omega)$ may be updated to $(t, \vec{r}', \Omega)$ if the combined finite network on $O \cup P_{B'}$ assigns $\vec{r}'$ to the new live ports and agrees with $\vec{r}$ on O . For every new live port p of B' and every $(R, D) \in \Omega$, if $\vec{r}'(p) = R$ then $D \in \tau_{B'}(p)$ is required. If the source type t contains $\forall R.D$, then (R, D) is included in Ω . No obligation is discarded because an intermediate relation temporarily leaves R ; the exact relation vector at every visited bag is carried explicitly.*

This repairs the earlier may-analysis. An obligation is never pruned merely because some composition entry is multivalued; it is evaluated against the relation actually recorded in the current bag network or shadow vector.

5.2 Patchwork abstraction

Definition 5.3 (Valid patchwork abstraction). *A valid patchwork abstraction for C_0 is a finitely branching tree whose nodes are labelled by patchwork bags of width at most $K(C_0)$ and satisfying:*

- (A1) *each bag network is a finite complete atomic RCC5 network with strong equality;*
- (A2) *adjacent bags agree exactly on their overlap positions, including types, relations, witness declarations that mention overlap positions, and shadow vectors;*
- (A3) *the equivalence classes generated by overlap maps are connected subtrees; these classes are the semantic occurrences of the final model;*
- (A4) *all existential restrictions in a port type have either a local witness in the same bag or a declared vertical parity request;*
- (A5) *all universal obligations are present as shadows and are checked against every live port introduced in every adjacent bag reached by shadow propagation;*
- (A6) *saturated side-context closure holds: any side/tree or side/side pair whose local network label is PP, PPI, or EQ is represented by a vertical stratum or shared identity port before the pair may be forgotten as residual frontier;*
- (A7) *residual frontier pairs are labelled only by DR or PO;*

(A8) *vertical eventuality requests are attached to a concrete support branch and, if the source occurrence is carried through an overlap, to the same overlap equivalence class.*

Theorem 5.4 (B: finite abstraction adequacy). *For every C_0 , saturated split-forest presentations of C_0 of width $K(C_0)$ are sound and complete for valid patchwork abstractions in the following sense.*

- (a) *Every saturated split-forest presentation induces a valid patchwork abstraction.*
- (b) *Every valid patchwork abstraction has a global abstract complete RCC5 interpretation that realizes all bag types, witnesses, universal restrictions, and vertical requests.*

Proof. For (a), label each decomposition bag by the restriction of the presentation's global RCC5 network, by the type of each occurrence, and by the selected witness/support declarations. Since the presentation is saturated, side/tree and side/side containment or identity pairs have already been verticalized or shared; residual frontier pairs are therefore DR or PO. Universal shadows are obtained by taking, for every source occurrence whose universal restrictions remain relevant outside the current bag, its type, its relation vector to the live ports, and the set of obligations induced by its universal formulas. Since these objects range over the finite set specified above, equal shadows can be merged. The merge is a congruence: the update into an adjacent bag depends only on $(t, \vec{r}, \Omega)$ and the finite adjacent bag network, not on the identity of the forgotten source.

For (b), form the set V of semantic occurrences as equivalence classes of local port occurrences under the overlap maps. For each bag B , the labelled network N_B is a finite complete satisfiable RCC5 network on the occurrences represented in that bag. Adjacent bags agree on overlaps by (A2), and the overlap classes are connected by (A3). By Theorem 2.1, the tree of bag networks has a countable abstract RCC5 completion ρ on V whose restriction to every bag is exactly N_B .

Concept names are interpreted according to the bag types; this is well-defined because bag types agree on overlaps. Boolean clauses follow from Hintikka consistency. Existential clauses are satisfied either by a same-bag witness, whose relation is read in N_B , or by a vertical parity witness whose support branch is realized in the tree and whose endpoint type contains the required concept. Universal clauses hold by the shadow condition: for a source occurrence x with $\forall R.D$, the obligation (R, D) is carried in every shadow/profile representing x outside its home bag. Whenever a target occurrence is introduced, the exact recorded relation from the source shadow to that target is checked; if it is R , then D is forced into the target type. Since every occurrence of V is introduced in some bag and every source's shadow is propagated along the unique connecting path in the decomposition tree, all targets are checked. Finally, strong equality is automatic: EQ occurs in a bag only on identical ports, and global identity is generated by overlaps, not by quotienting distinct vertices. Hence the interpretation realizes the abstraction. $\square$

Remark 5.5 (Why this repairs transformer coherence). *The proof of Theorem 5.4 never defines $\rho(x, y)$ by an interface route. It therefore has no least-common-ancestor versus higher-triple coherence problem. Pair and triple consistency is supplied by the finite bag networks and the RCC5 patchwork theorem. The diagnostic concept*

$$\exists \text{PP} . (A \sqcap \neg B \sqcap \forall \text{PP} . \neg B \sqcap \forall \text{PPI} . \neg B \sqcap \forall \text{PO} . \neg B) \sqcap \exists \text{PP} . B$$

is rejected because any patchwork bag or amalgam containing the shared split child and the two superpart witnesses must satisfy $\text{Comp}(\text{PPI}, \text{PP}) = \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}\}$, so DR is not available, and the remaining four relations are exactly those forbidden by the first witness type.

6 The automaton

Let Σ_{C_0} be the finite set of all patchwork bags of width at most $K(C_0)$: finite port sets, finite complete RCC5 networks, type assignments, overlap maps, witness declarations, shadows, and request declarations. Finiteness follows from finiteness of Rel , Typ , $\text{Cl}(C_0)$, and the bound $K(C_0)$.

Definition 6.1 (Two-way alternating parity automaton $\mathcal{A}_{C_0}$). *The automaton $\mathcal{A}_{C_0}$ runs on ranked trees over Σ_{C_0} . Its states are of six kinds.*

- (Q1) *local-bag states checking finite RCC5 consistency, type consistency, overlap agreement, and saturation closure;*
- (Q2) *existential-witness states checking local witnesses and spawning vertical request states for non-local PP/PPI requests;*
- (Q3) *universal-shadow states carrying a shadow profile and moving two-way through the decomposition tree to check every introduced target;*
- (Q4) *equality/identity states checking that overlap maps generate connected occurrence classes and that no distinct local ports are labelled EQ;*
- (Q5) *patchwork states checking that adjacent bag networks agree on their overlaps;*
- (Q6) *parity-request states for vertical eventualities.*

The transition relation is finite because every state mentions only a bounded port number, a type in Typ , a relation in Rel , a formula in $\text{Cl}(C_0)$, or a shadow profile from the finite shadow set.

The parity condition is a standard generalized Buchi condition converted to parity. For each vertical eventuality kind $e = (R, D)$ with $R \in \{\text{PP}, \text{PPI}\}$, the automaton creates a request state when it sees $\exists R.D$ without a same-bag witness. The request follows the declared support branch. It is fulfilled at the first occurrence whose recorded relation to the source is R and whose type contains D . If the request crosses an overlap, the source occurrence is the same overlap-equivalence class; there is no possibility of deferring forever by bouncing between semantic equality mates, because there are no nontrivial equality mates. Generalized Buchi acceptance requires every request generated infinitely often to be fulfilled infinitely often along the support run; a standard indexing of the finitely many request kinds gives a finite parity index.

Lemma 6.2 (Automaton soundness). *If $\mathcal{A}_{C_0}$ accepts an input tree, then the input tree is a valid patchwork abstraction for C_0 with some root occurrence whose type contains C_0 .*

Proof. The local-bag, overlap, identity, and patchwork states enforce (A1)–(A3). Existential states enforce (A4). Universal-shadow states enforce (A5): the automaton carries the full relation vector of the source shadow and checks the target type whenever the recorded relation is one of the source’s universal guards. Saturation states enforce (A6)–(A7). Parity-request states enforce (A8). The initial condition requires a root port with C_0 in its type. Hence all clauses of Definition 5.3 hold. $\square$

Lemma 6.3 (Automaton completeness). *Every valid patchwork abstraction for C_0 is accepted by $\mathcal{A}_{C_0}$.*

Proof. Choose the accepting run that follows the witnesses, shadows, overlaps, and support branches recorded in the abstraction. All local checks succeed by validity. Universal shadow runs succeed because the abstraction's shadow updates carry the exact relation vector into every adjacent bag. Every vertical eventuality request has a declared support branch reaching a witness of the required type, so the generalized Buchi condition is satisfied. Thus the parity run is accepting. $\square$

Theorem 6.4 (C: automaton correctness). *For every concept C_0 ,*

$$L(\mathcal{A}_{C_0}) \neq \emptyset \iff C_0 \text{ has a valid patchwork abstraction.}$$

Proof. Immediate from Lemmas 6.2 and 6.3. $\square$

7 Decidability

Theorem 7.1 (Decidability over abstract RCC5 frames). *Concept satisfiability for $\mathcal{ALCI}_{\text{RCC5}}$ over abstract complete atomic RCC5 frames with strong equality is decidable.*

Proof. Given C_0 , compute $\text{Cl}(C_0)$, the finite type set Typ , the finite width bound $K(C_0)$, and hence the finite alphabet Σ_{C_0} of patchwork bags. Construct $\mathcal{A}_{C_0}$. By Theorems 4.5, 5.4, and 6.4, C_0 is satisfiable iff $L(\mathcal{A}_{C_0})$ is nonempty. Emptiness for finite two-way alternating parity tree automata is decidable. Therefore satisfiability is decidable. $\square$

8 Expanded construction details

This section spells out the finite objects used above. The point is to make explicit where the previous route-dependent transformer machinery has been removed.

8.1 From a generated model to bags

Given a witness-generated model $\mathcal{I} \upharpoonright W$, choose for each generated occurrence x a home bag B_x . The home bag contains x , all selected same-step witnesses for existential formulas in $\text{tp}(x)$ that are not handled by a vertical request, and the bounded vertical boundary needed by the saturated side-context construction. If a selected witness y lies in a side context of x and some boundary point b has $\rho(y, b) \in \{\text{PP}, \text{PPI}, \text{EQ}\}$, then the bag is enlarged by the threshold position that witnesses that containment or identity interaction. If the threshold itself creates new side obligations, the rank decreases because it belongs to a proper subformula of the formula that generated y . Hence this recursive enlargement stops.

For multiple selected superparts, the same semantic occurrence is not duplicated as two RCC5-distinct points. Instead, it is represented by local ports in several bags joined by overlap maps. The equivalence class of those ports is one occurrence in the final tree decomposition. The connectedness condition on overlaps is the replacement for the old typed-EQ quotient. Thus the statement

$$p \text{ in } B \text{ and } q \text{ in } B' \text{ represent the same element}$$

means that p and q are in the same overlap-connected component; it does not mean that there are two distinct elements with an EQ edge between them.

8.2 The finite alphabet

Fix $K = K(C_0)$. The alphabet Σ_{C_0} is obtained by enumerating all tuples

$$(P, N, \tau, \mu, W, U, \Gamma)$$

with $P \subseteq \{1, \dots, K\}$, $N : P^2 \rightarrow \text{Rel}$, $\tau : P \rightarrow \text{Typ}$, μ a finite family of partial injective overlap maps to the parent and to the K possible child slots, W a set of witness declarations, U a set of shadow profiles, and Γ a set of vertical request declarations. A tuple is retained in the alphabet only if the following finite tests pass.

- (L1) $N(p, p) = \text{EQ}$ and $N(p, q) \neq \text{EQ}$ for $p \neq q$.
- (L2) $N(q, p) = N(p, q)^{-1}$.
- (L3) For all $p, q, r \in P$, $N(p, r) \in \text{Comp}(N(p, q), N(q, r))$.
- (L4) The type assignment τ is Hintikka-consistent.
- (L5) Every local existential witness declaration for $\exists R.D \in \tau(p)$ names a port $q \in P$ with $N(p, q) = R$ and $D \in \tau(q)$.
- (L6) Every nonlocal vertical request for $\exists R.D$ has $R \in \{\text{PP}, \text{PPI}\}$ and records which support branch it follows.
- (L7) For every universal formula $\forall R.D \in \tau(p)$, the shadow set contains the source obligation (R, D) for the occurrence represented by p whenever p is not kept live in the adjacent bag being checked.
- (L8) Saturation holds: if a side-context pair has label PP , PPI , or EQ , the tuple also contains the corresponding vertical stratum or overlap identification; otherwise it is not admitted as a residual frontier tuple.

The alphabet is finite because all components range over finite sets. Importantly, the alphabet does not contain a function that attempts to compute a far-away pair relation by iteration. It contains only finite complete networks on finite bags and finite summaries of obligations.

8.3 Overlap and running-intersection checks

For an edge between a parent bag B and a child bag C , the input label supplies a partial injection

$$\mu_{B,C} : P_B \rightarrow P_C.$$

The local transition checks:

$$\begin{aligned} \tau_C(\mu(p)) &= \tau_B(p), \\ N_C(\mu(p), \mu(q)) &= N_B(p, q) \end{aligned}$$

for all p, q in the domain of μ . The automaton also checks that each local port is introduced once and forgotten along a connected subtree. This can be expressed by three finite status bits for every port: introduced here, continued to a child, or forgotten here. Since the underlying input is a tree, these local bits enforce the running-intersection property for each overlap-equivalence class.

This is the exact point at which equality-port congruence is repaired. The old proof needed to show that two distinct copies had the same relation to every third occurrence. Here there are no two distinct copies. There is one global occurrence whose local manifestations form a connected subtree. All bag networks containing that occurrence agree on its relations to all other ports in the same bag by the overlap checks and patchwork agreement.

8.4 The patchwork construction in detail

Let T be an accepted labelled input tree. Let V_T be the quotient of all local port occurrences (t, p) by the equivalence relation generated by the overlap maps. For every node t , the bag label gives a finite complete network N_t on the finite set $P_t/\sim$ of global occurrences represented in that bag. If two bags share an occurrence, their networks agree on all relations inside the overlap.

For a finite set $F \subseteq V_T$, choose finitely many bags whose union contains F and close this set of bags under paths in the input tree. The result is a finite subtree of bags. Repeatedly apply the two-bag patchwork theorem from the leaves inward. This gives a satisfiable complete RCC5 network on the union of all occurrences in those bags, hence on F . Therefore every finite subset of V_T has at least one complete RCC5 labelling extending all bag networks. Since Rel is finite, the product space $\text{Rel}^{V_T \times V_T}$ is compact in the product topology. The closed set of labellings satisfying each finite bag constraint and each finite composition constraint has the finite intersection property, hence is nonempty. Any element of this intersection is the global relation ρ_T .

This construction proves three facts that were previously only asserted:

- (P1) ρ_T is well-defined globally, because it is chosen as one complete labelling of all pairs, not computed by a path.
- (P2) Every global triple satisfies the RCC5 composition table, because each finite triple is contained in some finite patchwork union and the compactness construction preserves all finite composition constraints.
- (P3) If a diagnostic composition is forced, such as $\text{PP}; \text{DR} = \text{DR}$ or $\text{PPI}; \text{PP} \subseteq \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}\}$, then every global completion respects that forced value or forced exclusion.

8.5 Universal propagation by exact relation vectors

A universal source x may be forgotten from a bag before a later target y is introduced. To ensure that $\forall R.D$ at x is not lost, the input carries a shadow profile for x . At a bag B , the shadow records the exact vector

$$(\rho_T(x, p))_{p \in P_B}$$

whenever x is not live. In an accepted input this vector is not inferred from a single composition step. It is part of the finite label and is checked for consistency with the parent and child bag networks. When a child bag introduces a new port q , the child label records $\rho_T(x, q)$ in the updated vector. If this relation is R and $\forall R.D \in \text{tp}(x)$, then D is required in the type of q .

This is stronger than may-analysis. A may-analysis would carry a set of possible relations and might drop an obligation if an intermediate step leaves R . Here the actual recorded relation at the target is checked. In an existential branch of the automaton the run guesses the vectors; soundness follows because the guessed vectors are part of the patchwork network and must be globally extendible. Completeness follows by choosing the vectors from an actual satisfying presentation.

8.6 Vertical eventualities and equality transfer

For $R \in \{\text{PP}, \text{PPI}\}$, a witness for $\exists R.D$ may lie arbitrarily far along a vertical branch. The automaton handles this by a parity request. A request state contains:

$$(R, D, p, \ell),$$

where p is the current manifestation of the source occurrence and ℓ is the support branch declaration. If the source occurrence passes through an overlap, the state updates p to the corresponding port

under the overlap injection. If the source is forgotten, its shadow is used to keep the relation vector. A request is fulfilled only when the run visits a port q with recorded relation R from the source and $D \in \tau(q)$.

Because occurrence identity is overlap identity, a request cannot be deferred by jumping between different equality mates. There are no equality mates outside the connected manifestations of the same global occurrence. The parity condition is therefore the standard one: for the finite set of request kinds (R, D) , require that each request kind that is pending infinitely often is fulfilled infinitely often. Converting the resulting generalized Buchi condition to a parity condition gives index at most $2 \cdot |\{(R, D) : R \in \{\text{PP}, \text{PPI}\}, D \in \text{Cl}(C_0)\}| + 1$.

9 Audit against the referee concerns

Concern	Repair in this manuscript
C1 transformer coherence	Pair/triple transformers are removed. Global coherence is obtained from the RCC5 patchwork property applied to complete bag networks agreeing on overlaps.
C2 multivalued composition freedom	Multivalued choices are made inside finite complete bag networks and globally amalgamated by patchwork; diagnostic triples such as $\text{PPI}; \text{PP}$ are checked in bags or finite patchwork unions, not inferred by route-dependent summaries.
C3 equality-port congruence	Nontrivial EQ quotienting is removed. Strong equality is identity of overlap-equivalence classes in a tree decomposition. Distinct local ports may never be labelled EQ.
M1 finite width	Numerical ad hoc recurrences are replaced by an effective type/mosaic bound: enumerate finite saturated context shapes up to the finite rank $ \text{Cl}(C_0) $.
M2 shadows	Shadows are explicit finite profiles $(t, \vec{r}, \Omega) \in \text{Typ} \times \text{Rel}^K \times 2^{\text{Rel} \times \text{Cl}(C_0)}$; updates factor through this profile.
M3 one-way sufficiency	The decision layer is stated as a two-way alternating parity tree automaton. No lossless one-way ancestor summary is assumed.
M4 universal propagation	Universal obligations carry exact relation vectors and are checked against the recorded relation at every introduced target. Obligations are not dropped by may-analysis.
m3 equality-eventualities	Requests are tied to overlap-equivalence classes, so they cannot be postponed by moving between distinct equality mates.

10 Limitations and scope

The theorem is for abstract complete atomic RCC5 frames satisfying the composition table. It does not assert decidability for topological RCC5 role semantics. The external algebraic input is the RCC5 patchwork property. If one wants a fully self-contained paper, Theorem 2.1 should either be proved in an appendix from a standard homogeneous representation of RCC5 or cited with a precise reference.

References

- [1] J. Renz and B. Nebel. Qualitative spatial reasoning using constraint calculi. Standard sources for RCC5/RCC8 composition and patchwork properties.
- [2] M. Bodirsky and S. Wolfl. Homogeneous representations and patchwork properties for region connection calculi.
- [3] M. Y. Vardi. Reasoning about the past with two-way automata, and standard emptiness results for two-way alternating parity tree automata.